



MOLPATH SIMULATOR

FLIGHT OPERATIONS MANUAL

Cockpit handbook · operational reference · molecular pathology training

EDITION	OPERATING MODEL	RUNTIME BASIS
English	3 COCKPITS + METHODS FOCUS	v2.6.0-rc1

OPERATING
PRINCIPLE

Read the situation. Protect the specimen. Choose the method. Interpret the evidence. Commit to a defensible decision.

OPERATIONS MANUAL · V0.6 · 25 AUG 2026

How this manual is organised

This handbook has two jobs. First, it explains how to configure and operate the simulator itself. Second, it remains an operational reference for the decisions each cockpit is meant to support: what to look at, when to stop, and what must be true before moving on.

SECTION	OPERATING AREA	CURRENT CASES	CORE FLIGHT PROFILE
SIM OPS	Simulator controls & navigation	All cases	Configure → select → navigate → complete → debrief / export
MTB	Patient care & tumour board	42 cases	Clinical question → specimen → diagnostic strategy → evidence → report → MTB decision
LAB	Laboratory · LIMS · QM	27 cases	Run context → QC event → reasoning → decision → root cause → CAPA → audit
RES	Research & translation	22 cases	Project idea → context → reasoning → hypothesis → PICO → design → methods → analysis → feedback
METHODS	Cross-domain focus layer	44 driver cases	Method fit → input validity → QC / analysis validity → inference limits → action

Runtime note: Methods focus is implemented as an independent cross-domain filter across MTB, LAB and RES. It is a second classification axis, not an independent fourth cockpit. Only cases in which methodology materially drives the correct resolution are included.

MANUAL LOGIC

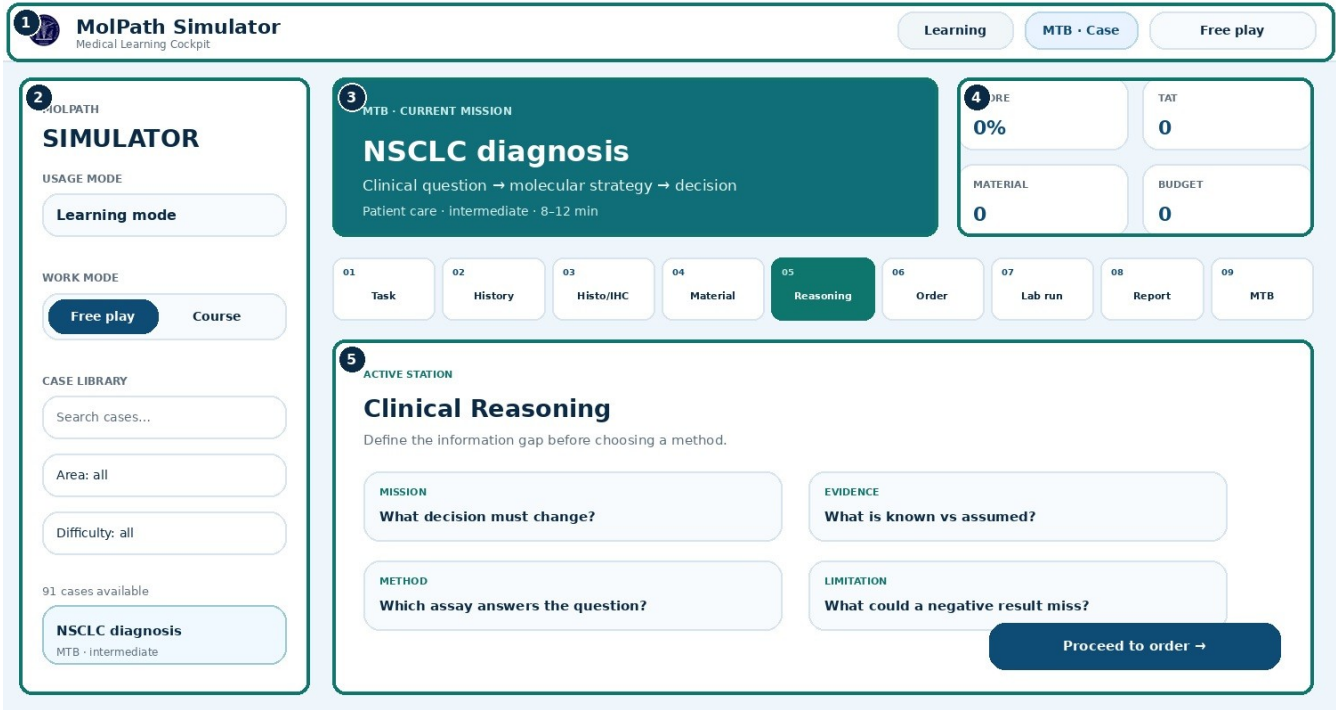
One simulator-operations layer, three operational cockpits and one orthogonal Methods Focus layer. Learn how modes, filters, instruments and reports behave first; then use the cockpit-specific decision doctrine.

DOCUMENT LANGUAGE

The operations manual is maintained once, in English. Simulator interface labels are referenced in their English form where available.

Cockpit anatomy

The simulator has a stable division of labour: navigation on the left/top, case mission and status centrally, and the active decision workspace in the main panel. The precise responsive arrangement changes on smaller screens; the operational meaning does not.



Schematic map of the current desktop cockpit. Responsive layouts relocate the same operational functions.

1 Top bar Work mode, case context and high-level navigation.	2 Navigation rail Usage mode, case library, filters and case switching.
3 Mission header Current case, domain, complexity and clinical/organisational frame.	4 Instruments Score, TAT, material and budget indicators where applicable.
5 Work area The active station: evidence, choices, methods, findings and decision controls.	

Four controls define how you enter a case

Before touching the case, separate four independent control layers. They answer different questions and can be combined without changing the identity of the case.

CONTROL LAYER	QUESTION	CURRENT CHOICES	WHAT IT CHANGES
OPERATING AREA	What kind of decision am I responsible for?	MTB · LAB · RES	Workflow, station sequence, landing decision and score context.
WORK MODE	How are cases accessed?	Free Play · Course	Whole library versus a guided learning path with progress and course reporting.
USAGE MODE	How much guidance should the simulator reveal?	Learning · Training · Assessment · Instructor	Hints, immediate feedback, live score and solution visibility.
DISCOVERY FILTERS	Which subset do I want now?	Search · Area · Methods Focus · Difficulty · Signature	Narrows the visible library; active filters intersect.

INDEPENDENT CONTROLS	Assessment is not a case type. Methods Focus is not a fourth cockpit. Course mode and Usage mode are independent - for example, Course + Assessment is a valid combination.
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60-SECOND FIRST FLIGHT

1	2	3	4	5	6	7
Choose work mode	Choose usage mode	Select area / filters	Open a case	Work through stations	Complete the landing	Debrief / report

DEFAULT	If you are unsure where to begin: Free Play + Learning mode + one operating area, with no additional filters. Add complexity only when it serves a learning objective.
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Free Play or Course?

Work mode determines how cases are accessed and how longitudinal progress is organised. It does not determine how much feedback you receive; that is controlled separately by Usage mode.

FREE PLAY • The complete current case library remains available. • Search, Area, Methods Focus, Difficulty and Signature filters can be combined freely. • You choose and switch cases individually. • Best for ad-hoc practice, exploration and targeted repetition. • No predefined sequence is required.	COURSE • One learning path and its case sequence become the active frame. • The dashboard tracks completed cases, average score and critical flags. • Continue / Next Case follows the course sequence. • Course Report 2.0 and course JSON become available once course progress exists. • Completed course progress is stored locally in the browser.
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WHERE TO CHANGE IT

UI PATH	Navigation rail → Work mode → Free Play / Course. In Course mode, choose the Learning path and use Start / Continue to enter the sequence.
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WHAT DOES NOT CHANGE

- The underlying case, correct decisions and cockpit workflow do not change because you entered it through a course.
- Usage mode remains independent. A course may be worked in Learning, Training, Assessment or Instructor mode.

RESET	Reset Case clears the active case state. Course Reset is separate and clears the stored progress of the selected learning path after confirmation.
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Choose how much the simulator should tell you

Usage mode controls guidance and disclosure. It is the same conceptual switch across MTB, LAB and RES, although the content shown by the cockpit differs.

BEHAVIOUR	LEARNING	TRAINING	ASSESSMENT	INSTRUCTOR
Didactic hints	High	Reduced	Minimal	Full
Immediate feedback	Yes	Yes	No	Yes
Score during case	Visible	Visible	Hidden	Visible
Model solution during case	No	No	No	Yes
Model solution after completion	Yes	Yes	Yes	Yes
Instructor Report 2.0	No	No	No	Yes
Best use	New content	Revision	Exam / self-check	Teaching / discussion

ASSESSMENT

Assessment suppresses immediate feedback, the live score and the model solution until the case is completed. It does not change the correct answer or the scoring logic.

INSTRUCTOR

Instructor mode exposes the model solution, expected decisions and scoring logic while the case is still open, and unlocks Instructor Report 2.0 for teaching preparation.

WHERE TO CHANGE IT

UI PATH

Navigation rail → Usage mode. The selected Usage mode is stored locally in the browser and remains active until you change it.

Find a case without changing its identity

Case-library controls answer a discovery question: which cases should be visible right now? They do not convert one cockpit into another. All active filters narrow the library together.

CONTROL	USE IT FOR	OPERATIONAL EFFECT
SEARCH	Free-text discovery by case content, identifiers and tags.	Narrows the visible list to matching cases.
AREA	Selecting the primary operating environment.	MTB / patient care, LAB / laboratory-QM or RES / research.
METHODS FOCUS	Finding cases in which methodology materially drives the resolution.	Second classification axis; can show all method drivers or one methodological domain.
DIFFICULTY	Targeting the intended complexity level.	Beginner · Intermediate · Advanced · Expert.
SIGNATURE	Restricting the library to curated representative / showcase cases.	Keeps only cases marked as Signature.

METHODS FOCUS DOMAINS

Molecular diagnostics · Preanalytics / analytics / QC · Research & experimental design · Bioinformatics & statistics · Digital pathology & AI

FILTER LOGIC	Example: MTB + Advanced + Signature + Molecular diagnostics returns only cases that satisfy all active criteria. Methods Focus remains an overlay; the case keeps its original MTB, LAB or RES workflow.
CAUTION	Switching to another case initializes a new active case state. Unfinished current-case choices are not a persistent course checkpoint; complete the case or export a session record before leaving if you need a record of that work.

Read the instruments as constraints, not decoration

The instrument cluster is cockpit-dependent. In MTB it models diagnostic resources; in LAB and RES the same display area is repurposed for the operational state of that cockpit.

COCKPIT	INSTRUMENT CLUSTER	HOW TO READ IT
MTB	Score · TAT · Material · Budget	Performance plus specimen, time and didactic-resource constraints.
LAB	Score · Run status · QC · Mode	Release / process state, QC event visibility and laboratory context.
RES	Score · Phase · Mode · Area	Research progress and the current simulation context.

MTB RESOURCE LOGIC

INSTRUMENT	CURRENT RUNTIME CALCULATION	INTERPRETATION
MATERIAL	Sum of the sections consumed by the selected tests, relative to the case limit.	Protect diagnostic reserve and avoid consuming tissue without decision value.
BUDGET	Sum of didactic resource points assigned to selected tests, relative to the case budget.	Resource awareness only - not real billing or reimbursement.
TAT	Longest selected test TAT, assuming parallel execution, plus 4 workdays per follow-up request.	The broadest path may not be the fastest; a later add-on is visible as delay.
SCORE	Performance generated from cockpit-specific decision domains.	Read the domain profile and any critical cap, not only the headline percentage.

EXAMPLE	A 5-day assay plus a 3-day assay selected together yields a 5-day base TAT, not 8 days. A later follow-up request adds 4 workdays in the current v2.6.0-rc1 runtime.
STEWARDSHIP	Optimisation does not mean "order less". It means maximising decision-relevant information while preserving tissue, time and a route for reflex or orthogonal testing.

Scores reward defensible decisions, not button completion

The simulator scores the quality of the reasoning path. Different cockpits use different score domains, and case-specific critical errors may restrict the achievable final result even when other decisions were strong.

MTB	LAB	RES
Clinical reasoning Method selection Material stewardship Biological interpretation Communication	Patient safety Process decision Documentation CAPA / audit readiness Communication	Clinical question Hypothesis / PICO Study design Analysis plan Translation / validation

HOW THE SCORE IS BUILT

- Case decisions add or remove credit in one or more relevant domains; the score is not a count of completed controls.
- Low-value, invalid or incomplete choices are penalised according to the decision they compromise - for example method fit, material stewardship, safety or study design.
- Reasoning Gates and other critical decisions can impose a case-specific score cap when a core question, validity requirement or safety boundary is missed.
- The runtime normalises the resulting performance to a percentage and shows the domain profile. Exact weights and cap thresholds are versioned and may be case-specific.
- Assessment mode hides the score during the case; it does not use a different scoring engine.

READ A SCORE	Use this order: headline percentage → domain profile → critical findings / cap reason → missed opportunities → targeted repeat. A single percentage without the domain pattern is intentionally incomplete.
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INSTRUCTOR VIEW	Instructor mode and Instructor Report 2.0 expose expected decisions and score logic for teaching. Learners should not need those details to complete a normal case defensibly.
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Move through stations, then commit before debrief

Station names and counts differ between MTB, LAB and RES, but the interaction pattern is stable: gather evidence, reason, make a defensible choice, commit, then compare only after the operational endpoint has been reached.

STEP	WHAT YOU DO	WHAT THE SIMULATOR DOES
1 · READ	Open the current station and inspect the presented context, assets, QC or project information.	Marks the station as visited and exposes the controls relevant to that phase.
2 · REASON	Separate evidence from assumptions and complete the Reasoning Gate when present.	Uses the gate as a prerequisite before downstream decisions in normal learner modes.
3 · SELECT	Choose methods, release action, study components or other case-specific decisions.	Tracks selections for feedback, resources and scoring.
4 · COMMIT	Proceed / run / submit once the decision is defensible.	Advances the case and records the consequences of the chosen strategy.
5 · LAND	Complete the final MTB, audit or research-feedback endpoint.	Finalises the case and unlocks the normal post-case feedback/reporting layer.
6 · DEBRIEF	Compare your path with expected reasoning, score and take-home messages.	Reveals the solution according to Usage mode and completion state.

DECISION GATE	Do not treat the gate as a formality. Its function is to make the information gap, risk or inference target explicit before downstream selections can make the case look deceptively complete.
RESET CASE	Reset Case reinitialises the active case choices and progress. It is separate from Course Reset and does not erase already stored course history.
ASSESSMENT	Immediate correctness feedback and the live score remain suppressed until completion; the station sequence itself remains the same.

Choose the output for the job

The simulator produces several outputs because a learner review, a teaching handout, an instructor briefing and a machine-readable session record are different products. Choose the report by purpose, not by filename.

OUTPUT	BEST FOR	AVAILABILITY	FORMAT / CONTENTS
CASE REPORT 2.0	Reviewing one completed case in detail.	After case completion; also available while open in Instructor mode.	Printable report with decisions, reasoning, score/domain findings, critical issues and expected solution.
TEACHING SHEET	Case discussion, debrief or a concise teaching handout.	After completion; also available while open in Instructor mode.	Compact case logic, correct path, traps, critical point, take-home messages and discussion questions.
INSTRUCTOR REPORT 2.0	Preparing or supervising teaching.	Instructor mode only; available while the case is open.	Learning goals, expected path, gate solution, score/cap logic, pitfalls, twist and teaching prompts.
COURSE REPORT 2.0	Reviewing performance across a learning path.	Course mode after at least one course case has been completed.	Progress, aggregate performance, domain pattern, critical issues, repeat targets and recommendations.
SESSION JSON	Machine-readable capture of the current case state.	Available from Actions during a session.	Case/mode state, visited stations, selections, follow-ups, decisions and score data. Not a human-readable report.
COURSE DATA JSON	Machine-readable course-progress data.	Course mode once course progress exists.	Stored course completion and performance data for reuse or analysis.

WHICH ONE?

Learner review → Case Report. Case discussion → Teaching Sheet. Teaching preparation → Instructor Report. Longitudinal course review → Course Report. Data capture / analysis → JSON.

PDF

Human-readable reports open in a report window and can be printed or saved as PDF using the browser print function. If the report window does not appear, allow pop-ups for the simulator.

Know what persists, what resets, and where controls move

The simulator deliberately keeps persistent course preferences separate from the active in-memory case. This matters when you reload, switch cases or work on a compact screen.

STATE	STORED WHERE?	PERSISTS AFTER RELOAD?
Language	Local browser storage	Yes
Usage mode	Local browser storage	Yes
Work mode	Local browser storage	Yes
Selected active course	Local browser storage	Yes
Completed course progress	Local browser storage	Yes
Unfinished current-case choices	Active session memory	No - reload or case switch starts a fresh case state
Exported session JSON	Downloaded file	Yes as a record; it is not an automatic resume function

BEFORE LEAVING	If unfinished work must be documented, export the session JSON before reloading or switching cases. The current runtime does not use that export as an automatic resume checkpoint.
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COMPACT / TABLET VIEW

CONTROL	WHAT IT OPENS
MENU	Navigation rail / case controls
OVERVIEW	Dashboard or course overview
SCORE	Score sheet / domain view
SELECTION	Selected-test cart when a cart exists

On screens up to the compact breakpoint, desktop panels are moved into these sheets rather than removed. Tables may scroll horizontally and content stacks vertically; the operational meaning of the controls does not change.

SESSION CONTROL	Reset Case clears the active case. Course Reset clears the selected course progress. These are intentionally separate actions.
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Six rules for every cockpit

RULE 01 Mission before method First define the decision the case must support. A technically possible test is not automatically a useful test.	RULE 02 Evidence before inference Separate what the case actually shows from what you infer. Assets and QC belong to the evidence layer.
RULE 03 Protect the specimen Material is a finite resource. Tissue consumption, sample quality and the option to preserve material are part of the decision.	RULE 04 Negative ≠ excluded A negative result only excludes what the method, specimen and sensitivity were capable of detecting.
RULE 05 Result ≠ action Translate a finding through biological meaning, evidence level, clinical context and limitations before recommending action.	RULE 06 Commit before debrief The simulator’s learning value depends on making a defensible decision before reading the final feedback.

The manual uses four operational callouts throughout:

NOTE	Orientation or explanatory context.
CAUTION	A common error that may invalidate or overstate the decision.
DECISION GATE	A checkpoint that must be passed before the next operational phase.
NO-GO	Do not advance until the stated problem has been resolved or explicitly accepted as a limitation.

Nine stations from task to tumour board

The current MTB/patient-care runtime uses nine stations. Think of them as five operational phases rather than nine isolated pages.

01	02	03	04	05	06	07	08	09
Task	History	Histology / IHC	Material	Clinical Reasoning	Order	Lab run	Report	MTB

PRE-FLIGHT 01–04 → DECISION GATE 05 → EXECUTION 06–07 → APPROACH 08 → LANDING 09

STATION	PRIMARY OUTPUT	DO NOT PROCEED IF...
01 Task	A one-sentence clinical decision question	You cannot state what decision the molecular workup is supposed to change.
02 History	Relevant diagnosis, stage, therapies and timing	Prior therapy or disease context is unknown but materially changes interpretation.
03 Histology/IHC	Tumour identity/phenotype and relevant morphology	The biological target of testing is uncertain or the available tissue is not characterised.
04 Material	Usability, tumour fraction, quality, reserve and constraints	Sample limitations make the planned test non-informative or unsafe for remaining material.
05 Reasoning	Prioritised diagnostic hypothesis and information gaps	You are ordering tests without a defined information gain.
06 Order	Minimal sufficient method strategy	The chosen method does not answer the biological question.
07 Lab run	Technically valid evidence with QC understood	QC or assay limitation prevents a valid biological conclusion.
08 Report	Structured molecular interpretation with limitations	Finding and clinical action are being conflated.
09 MTB	Concrete, qualified recommendation / next step	Uncertainty, evidence level or required follow-up is hidden from the final statement.

Build the situation before you order anything

The first four stations are not “background reading”. They define whether a later molecular result can be interpreted at all.

MISSION 01 · TASK Restate the mission in one sentence: Which immediate clinical decision should the molecular workup support?	CONTEXT 02 · HISTORY Extract only context that can change test selection or interpretation: tumour entity, stage, prior/current therapy, response/progression and timing.
IDENTITY 03 · HISTOLOGY / IHC Confirm the biological object you are testing. Use morphology and IHC to anchor lineage, phenotype and relevant differential questions.	RESOURCE 04 · MATERIAL Treat the specimen as an operating constraint: tumour fraction, necrosis, fixation/decalcification, nucleic-acid quality, block reserve and alternative material.
CAUTION	Do not let an attractive biomarker question outrun the specimen. A perfect assay on the wrong or non-representative material is still a poor decision.

PRE-FLIGHT CLEARANCE QUESTIONS

- ☐ What is the decision endpoint: diagnosis, therapy selection, resistance mechanism, hereditary implication or additional testing?
- ☐ What information is known versus assumed?
- ☐ Is the available sample representative and technically usable for the intended method?
- ☐ What must be preserved for reflex testing, orthogonal confirmation or future diagnostics?

CLEARANCE	Proceed to Clinical Reasoning only when the mission, disease context, biological target and specimen constraints can each be stated explicitly.
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Clinical Reasoning is the point of commitment

The Clinical Reasoning station separates passive review from active strategy. Before selecting tests, define the information you need and what you would do with each possible result.

STATE	QUESTION	INTERPRETATION	ACTION
GREEN	Will the answer change management or a required next step?	Clear information gain.	Order / proceed.
AMBER	Could the result be useful, but only under a condition?	Reflex, optional or context-dependent value.	Define trigger first.
RED	Is the test broad, redundant or unable to answer the question?	Low-value or misleading information.	Do not order.

A strong reasoning gate contains four explicit elements:

- Diagnostic hypothesis or clinical information gap.
- Expected decision impact of a positive, negative or indeterminate result.
- Methodological constraints that affect sensitivity or interpretability.
- A stopping rule: when is the evidence sufficient to move to reporting/MTB?

DECISION GATE	If you cannot explain why a test result would change the next step, the order is not yet justified.
NO-GO	Do not use “broad molecular profiling” as a substitute for an undefined clinical question.

Choose the method by the biological question

The order station is an information-design problem. Match the alteration class and decision requirement to the method while balancing tissue, turnaround time and uncertainty.

BIOLOGICAL QUESTION	METHOD FAMILY	READ THE LIMITATION
Sequence variant / small indel	DNA NGS; targeted PCR when the question is narrow and predefined	Coverage, VAF, FFPE artefact, breadth vs sensitivity
Fusion / splice event / transcript	RNA-based fusion testing; targeted RNA; FISH as orthogonal/rapid option in defined settings	RNA quality, expression, partner coverage, DNA-negative ≠ fusion-negative
Protein / phenotype	IHC	Antibody/assay performance, controls, scoring, heterogeneity
Amplification / rearrangement	ISH/FISH and/or validated NGS/CNV approach	Tumour content, signal interpretation, assay-specific thresholds
MMR / MSI context	MMR IHC and MSI testing as complementary strategies where appropriate	Discordance, preanalytics, downstream hereditary context
Methylation question	Targeted methylation assay or classifier depending on the question	Local marker vs genome-wide classification
Liquid biopsy	ctDNA testing when clinically and biologically appropriate	Shedding and sensitivity; negative result requires context
METHODS FOCUS	Method-driven MTB cases now sit inside the cross-domain Methods Focus filter. Historical MET identifiers remain for compatibility, but eligibility is curated by whether assay choice, validity, sensitivity, coverage, artefact recognition or another methodological constraint materially determines the resolution.	

ORDER OPTIMISATION

Aim for the smallest strategy that is still sufficient. Optimisation is not “order less”; it is maximise decision-relevant information per unit of tissue, time and uncertainty.

Read validity before meaning

A molecular result is only interpretable after the run and assay context are understood. The simulator deliberately places QC, technical limitations and report language between the test order and the final MTB decision.

1 VALID? Were controls, coverage, sample quality and assay-specific QC acceptable?	2 DETECTED? What alteration or biomarker was actually observed, at what quality/confidence?	3 EXCLUDED? What can the assay reasonably exclude — and what remains outside its reach?	4 RELEVANT? What is the biological and clinical significance in this exact tumour and treatment context?
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REPORT READING ORDER

- Primary finding(s) and assay result.
- Analytical quality and sample-specific limitations.
- Biological interpretation and classification.
- Evidence supporting clinical relevance.
- Required reflex, confirmation, re-biopsy or hereditary follow-up.

CAUTION	A negative report is not a universal statement that the biology is absent. Phrase the conclusion within assay coverage, sensitivity, specimen quality and tumour fraction.
NO-GO	Do not convert a VUS, low-confidence artefact or technically unsupported negative into a treatment statement.

Translate evidence into a tumour-board statement

The final station is not a repetition of the report. It is a decision synthesis: what the evidence means now, how certain that conclusion is, and what should happen next.

1 · CONTEXT	In [tumour / clinical setting]...
2 · EVIDENCE	the current molecular workup demonstrates / does not demonstrate...
3 · INTERPRETATION	This is clinically relevant because...
4 · LIMITATION	Interpretation is limited by...
5 · ACTION	Therefore: recommend / consider / repeat / obtain / refer...

LANDING CHECKS

- ☐ The recommendation answers the original task, not a different question discovered later.
- ☐ Actionability is separated from biological interest.
- ☐ Evidence level and tumour-specific applicability are explicit.
- ☐ Uncertainty and assay limitations are visible, not buried.
- ☐ Any germline implication, reflex test or re-biopsy requirement is handed off clearly.

LANDING	A strong MTB statement is short enough to read aloud, specific enough to act on, and qualified enough to remain scientifically defensible.
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Only after landing should you use the debrief to compare your strategy with the simulator's expected reasoning, score and take-home messages.

When the normal flight profile breaks

ABNORMAL CONDITION	OPERATING RESPONSE	AVOID
INSUFFICIENT MATERIAL	Stop and determine whether a smaller/alternative assay, macrodissection, new block, re-biopsy or alternative specimen can answer the mission.	Do not consume the last material simply to complete the nominal panel.
QC FAILURE	Separate run validity from patient result. Repeat, investigate or withhold interpretation according to the failure mode.	Do not “rescue” an invalid result by narrative confidence.
DISCORDANT RESULTS	Check sample identity, preanalytics, assay orthogonality, tumour heterogeneity and biological plausibility before choosing a preferred result.	Do not average incompatible results into a false consensus.
VUS / UNCERTAIN FINDING	Report uncertainty, review evidence and avoid therapeutic overcalling. Use orthogonal confirmation only if it resolves a real decision.	Do not convert database presence or computational prediction into actionability.
POSSIBLE GERMLINE SIGNAL	Recognise the possibility, separate tumour finding from germline diagnosis and define the appropriate genetics pathway.	Do not declare inherited status from tumour-only sequencing.
NEGATIVE BUT SUSPICION REMAINS	Ask whether the specimen and method had adequate sensitivity for the suspected alteration class; consider a complementary method or new material.	Do not equate “not detected” with “biologically excluded”.
PRIORITY	When evidence and process disagree, protect validity first. A delayed defensible decision is preferable to a fast unsupported one.	

Take-off checklist

Use before committing to the diagnostic strategy. This is a pre-decision checklist, not a substitute for the full case workflow.

MISSION ☐ Clinical question understood ☐ Tumour entity, stage and decision context identified ☐ Previous/current therapy reviewed where relevant ☐ Immediate decision to be supported is clear	SPECIMEN ☐ Relevant specimen identified ☐ Tumour content and representativeness considered ☐ Material / preanalytic limitations recognised ☐ Reserve or alternative specimen considered if required
TEST STRATEGY ☐ Mandatory biomarkers / information gaps identified ☐ DNA / RNA / IHC / FISH / PCR requirements considered ☐ Method chosen for the biological question ☐ Tissue consumption, TAT and information gain considered	BEFORE COMMITTING ☐ QC requirements anticipated ☐ Negative-result limitations considered ☐ Evidence / actionability threshold defined ☐ Potential hereditary implications and reflex testing considered

TAKE-OFF
CLEARANCE

Proceed when the mission is explicit, the specimen is fit for purpose, the selected method answers the question, and you know how a positive, negative or indeterminate result would change the next step.

CASE ID

DATE

CLEARED ☐

Manual modules: MTB Operations · LAB Operations · RES Operations · Methods Focus. The same visual language and decision-gate doctrine now spans all three cockpits and the cross-domain methodological layer.

Quality is a release decision, not a checkbox

The LAB cockpit simulates the point at which technical evidence, process control and patient safety converge. The objective is not merely to “find the error”; it is to decide what may be released, what must be held, how the problem is investigated, and what makes closure audit-ready.

CURRENT CASES	RUNTIME PROFILE	PRIMARY PRIORITY
27 LAB	7 stations	Patient safety

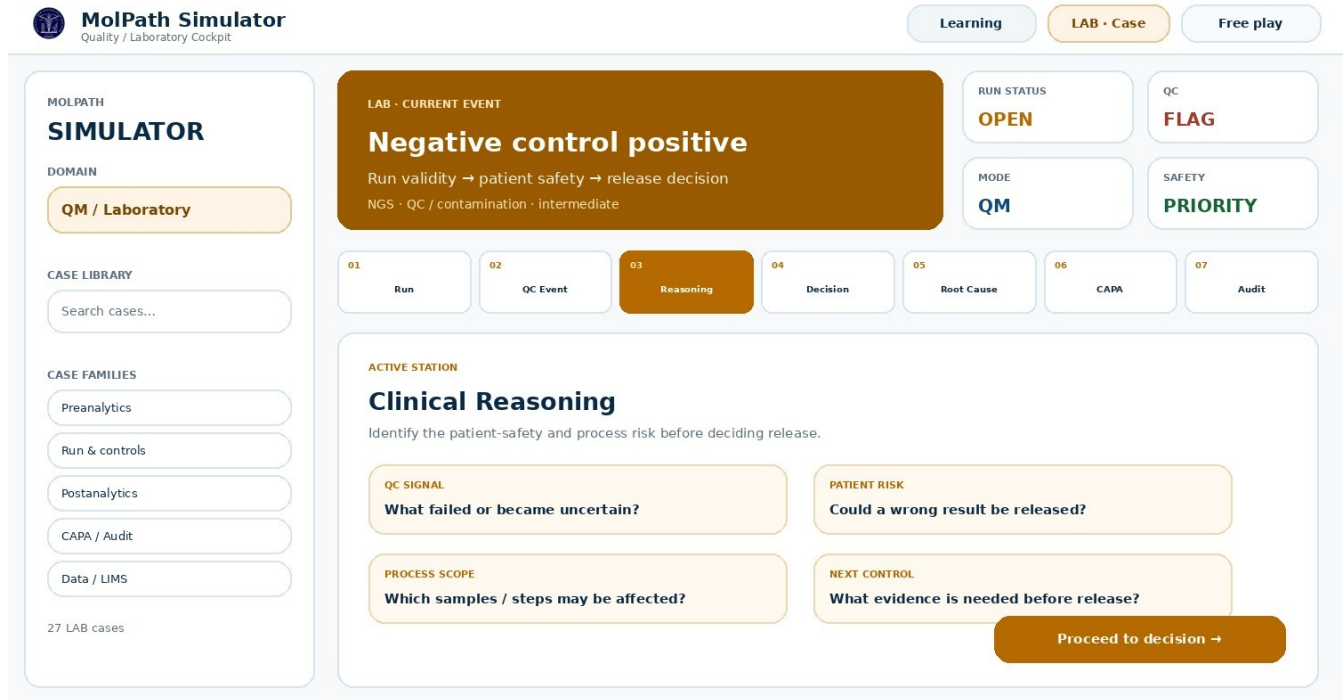
OPERATING PRINCIPLE	Validity first. Release second. Root cause third. CAPA only closes when prevention and effectiveness are demonstrable.
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CASE FAMILIES

LAB / PRE Preanalytics & specimen management Identity, sample suitability, fixation, decalcification, liquid-biopsy handling and material protection.	LAB / RUN Analytical run & controls Control failures, contamination, inhibition, lot verification and run-release logic.
LAB / POST Postanalytics & bioinformatics Pipeline changes, database versions, report classification, coverage gaps and traceability.	LAB / QM Documentation · audit · CAPA Deviation handling, SOP/version control, training records, root cause, effectiveness and audit readiness.
LAB / DIGITAL Digital pathology · data · LIMS Image quality, local validation, privacy, IT security, ransomware and downtime workflows.	LAB / SIM Planspiel escalation Complex scenarios in which process continuity, patient safety and recovery planning must be managed in parallel.

The quality cockpit

The LAB interface retains the simulator's common navigation shell but replaces clinical-test ordering with a release-and-control workflow. Status indicators become operational: run status, QC state, mode and safety risk.



Schematic representation of the LAB cockpit based on the current seven-station runtime.

COCKPIT SHIFT

In MTB, the central question is "Which evidence do I need?". In LAB, it becomes "Can this process/result be trusted and released - and what must happen if it cannot?".

Seven stations from event to audit closure

01	02	03	04	05	06	07
Run	QC Event	Clinical Reasoning	Decision	Root Cause	CAPA	Audit

SITUATION 01–02 → DECISION GATE 03 → RELEASE CONTROL 04 → INVESTIGATION 05 → SYSTEM RESPONSE 06 → CLOSURE 07

STATION	PRIMARY OUTPUT	DO NOT PROCEED IF...
01 Run	Run / process context and operational constraints	You do not know what process, specimens, controls or urgency are involved.
02 QC Event	Defined signal, deviation or uncertainty	The event has not been separated from normal technical variation or documentation noise.
03 Reasoning	Explicit patient-safety / quality risk and affected scope	You are choosing release or CAPA actions before stating what could be wrong.
04 Decision	Release, partial release, hold or repeat decision	The release decision cannot be justified from the evidence and risk.
05 Root Cause	Structured investigation plan and plausible causal hypotheses	The investigation only repeats the test without examining the process.
06 CAPA	Deviation + containment + correction + prevention + effectiveness plan	Action addresses the result but not recurrence or system learning.
07 Audit	Traceable closure and demonstrable effectiveness	The chain from event to evidence, action, responsibility and closure is incomplete.

Define the risk before you decide the release

The most important LAB error is premature action: repeating a run, editing a report or releasing “plausible” samples before the safety and process risk has been defined.

SITUATION 01 · RUN CONTEXT What assay or process is running? Which specimens, controls, batches, instruments, software versions and personnel are involved? What is the urgency?	SIGNAL 02 · QC EVENT What changed the validity state? Control signal, coverage warning, contamination, inhibition, documentation gap, pipeline change, image quality, data event or specimen problem?
RISK 03 · CLINICAL REASONING What wrong patient result or process outcome could occur? Which samples, markers, timepoints or downstream decisions might be affected?	BOUNDARY SCOPE BEFORE ACTION Distinguish local, batch-wide and system-wide uncertainty. Scope determines whether partial release is even thinkable.

REASONING GATE QUESTIONS

- ☐ What is the patient-safety or quality risk if I release now?
- ☐ Which evidence is trustworthy and which evidence has become uncertain?
- ☐ What is the smallest defensible affected scope - and what evidence supports that boundary?
- ☐ What additional observation, repeat or traceability check would materially change the release decision?

DECISION GATE

Do not proceed to the release console until you can state the risk, the affected scope and the evidence needed to reduce uncertainty.

Release, partial release, hold or repeat

A LAB decision is risk-based, not convenience-based. Turnaround pressure matters operationally, but it does not restore analytical validity.

STATE	WHEN DEFENSIBLE	MINIMUM EVIDENCE	DEFAULT
RELEASE	No relevant validity concern remains.	Controls/process trace support the intended interpretation.	Proceed
PARTIAL	A clearly independent subset is demonstrably unaffected.	Defined scope + documented independence + no hidden shared failure mode.	Exceptional
HOLD	Validity or patient-safety risk is unresolved.	Uncertainty itself is material to release.	Preferred under uncertainty
REPEAT	A repeat can resolve a defined hypothesis or restore valid evidence.	Repeat starts from the appropriate process step and is interpreted with the original event.	Conditional
CAUTION	“The patient result looks plausible” is not evidence of independence from a failed control or shared process error.		
NO-GO	Do not release an affected result because the VAF, morphology, previous history or clinical expectation makes it look believable.		

PARTIAL RELEASE TEST

Partial release is defensible only when the unaffected subset is technically independent of the failure mode and that independence can be documented. If the boundary is assumed rather than demonstrated, treat the subset as affected.

Root cause is a hypothesis-driven investigation

Root-cause analysis is not a ritual list of possible mistakes. Start from the observed failure pattern, build plausible causal hypotheses, then choose evidence that discriminates between them.

1	2	3	4	5
PATTERN What exactly failed, where, when and with what distribution?	SHARED PROCESS Which specimens or results share plate position, reagent, lot, instrument, operator, software or work zone?	HYPOTHESIS Contamination? inhibition? sample mix-up? reagent/lot? pipeline/version? documentation? preanalytics?	DISCRIMINATING TEST Which check or repeat would separate the leading hypotheses?	SCOPE UPDATE Does new evidence shrink or expand the affected population?

EVIDENCE SOURCES

- Plate / batch layout and adjacency patterns.
- Positive and negative controls, raw signal and assay-specific QC.
- Reagent lots, preparation history and storage conditions.
- Pre-PCR / post-PCR zoning, pipetting sequence and environmental/process contamination risk.
- LIMS / ELN / instrument metadata, sample identity and traceability chain.
- Pipeline, database, software and configuration versions where postanalytics are implicated.
- Repeat testing from the process step appropriate to the hypothesis - not automatically from the last step.

CAUTION

“Repeat was negative” can correct a result without explaining why the original process failed. Repeat evidence belongs inside the root-cause model; it does not replace one.

CAPA is the bridge from event to prevention

The simulator deliberately treats CAPA as more than “repeat and document”. A complete response links the observed event to containment, root cause, corrective action, prevention and an effectiveness check.

01 · DEVIATION Record what happened, when it was detected, affected scope and immediate status.	02 · CONTAINMENT Protect patients and the process now: hold, segregate, stop release, preserve evidence.
03 · ROOT CAUSE Document the causal conclusion and the evidence supporting it.	04 · CORRECTIVE Restore the affected process/result: repeat, repair, reanalyse, retrain or reconfigure as justified.
05 · PREVENTIVE Change the system so the same mechanism is less likely to recur.	06 · EFFECTIVENESS Define how and when you will verify that the intervention actually worked.
CAPA QUALITY TEST	For every action, ask: Which failure mechanism does this action address, who owns it, when is it due, and what observation will prove effectiveness?

WEAK CLOSURE LANGUAGE

- “Run repeated.”
- “Staff reminded.”
- “Issue discussed.”
- “No further problems observed.”

These statements may describe activity, but they do not by themselves demonstrate cause, prevention, responsibility or effectiveness.

Make the evidence chain reconstructable

Audit readiness is the final operational test: could an independent reviewer reconstruct the event, the decision, the evidence, the actions and the closure without relying on memory or informal explanation?

1 · EVENT	What happened? Raw observation, date/time, specimen/run/process identifier.
2 · DECISION	What was held, released, repeated or recalled - and why?
3 · INVESTIGATION	Which hypotheses were tested and what evidence supported the final root cause?
4 · ACTION	Which corrective/preventive actions were implemented, by whom and under which controlled document/version?
5 · EFFECTIVENESS	How was recurrence risk checked and what objective evidence closed the CAPA?

AUDIT-READY CLOSURE CHECKS

- ☐ Sample/run identity and traceability chain are intact.
- ☐ SOP, software/pipeline, reagent/lot and training versions are identifiable where relevant.
- ☐ The release decision is linked to the evidence available at that time.
- ☐ Deviation and CAPA records contain ownership, deadlines and status.
- ☐ Effectiveness is demonstrated with evidence rather than assumed from elapsed time.
- ☐ Any patient-impact action (repeat report, recall, amended report, notification) is traceable.

LANDING	LAB landing occurs when the process is safe to operate again and the reason for that conclusion is independently reconstructable.
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High-risk situations and default responses

ABNORMAL CONDITION	OPERATING RESPONSE	AVOID
POSITIVE NEGATIVE CONTROL	Hold the run/affected results; assess contamination scope; review layout, workflow, reagents and repeat strategy.	Do not ignore a low-level signal because the patient calls look plausible.
CONTROL / QC FAILURE	Separate instrument/assay validity from patient interpretation; define whether failure is global or analyte/sample-specific.	Do not rescue an invalid run by narrative qualification alone.
PIPELINE / VERSION CHANGE	Establish change control, validation/verification status, impact on already generated results and reanalysis need.	Do not treat undocumented software change as a clerical issue.
TRACEABILITY GAP	Stop release where identity or evidence chain is uncertain; reconstruct source data, LIMS/ELN trail, SOP/version and operator context.	Do not fill gaps from memory after the fact.
DATA / IT EVENT	Protect confidentiality, integrity and availability; invoke backup/downtime/recovery procedures and preserve audit trail.	Do not restore service without verifying data integrity.
PARTIAL RELEASE PRESSURE	Require demonstrable technical independence of the subset and explicit documentation of residual risk.	Do not let urgency redefine the affected scope.
PRIORITY	Patient safety and process validity outrank TAT comfort. A delayed defensible release is preferable to a rapid result whose validity cannot be defended.	

Take-off checklist

Use before committing to a release / hold / repeat decision. The checklist is deliberately front-loaded: once an invalid result is released, later CAPA cannot undo the patient-safety exposure.

SITUATION ☐ Run / process and case context understood ☐ Relevant samples, controls, lots, versions and urgency identified ☐ Observed QC event separated from assumptions ☐ Potential downstream patient decision understood	RISK & SCOPE ☐ Patient-safety / quality risk stated explicitly ☐ Affected scope defined and evidence-based ☐ Shared process dependencies considered ☐ Residual uncertainty acknowledged
RELEASE DECISION ☐ Release / partial release / hold / repeat option justified ☐ Partial release independence demonstrated if used ☐ Repeat starts from the appropriate process step ☐ TAT pressure not used as evidence of validity	SYSTEM RESPONSE ☐ Deviation / traceability requirements identified ☐ Root-cause hypotheses linked to evidence ☐ CAPA includes correction + prevention + owner + deadline ☐ Effectiveness check defined before closure

TAKE-OFF
CLEARANCE

Proceed when you can defend the validity state, affected scope, release decision and the evidence required for safe closure.

CASE / RUN _____

DATE _____

CLEARED ☐

Next section: Methods Focus. MTB, LAB and RES form the three stable operational cockpits; the final manual section overlays method-driven cases across all three.

A technology is not a research question

The RES cockpit trains the decisions that turn an interesting observation, available platform or project idea into a study that is interpretable, reproducible and translationally credible. The central discipline is to define the inference before generating the data.

CURRENT CASES	RUNTIME PROFILE	SCORE DOMAINS
22 RES	9 stations	5 dimensions

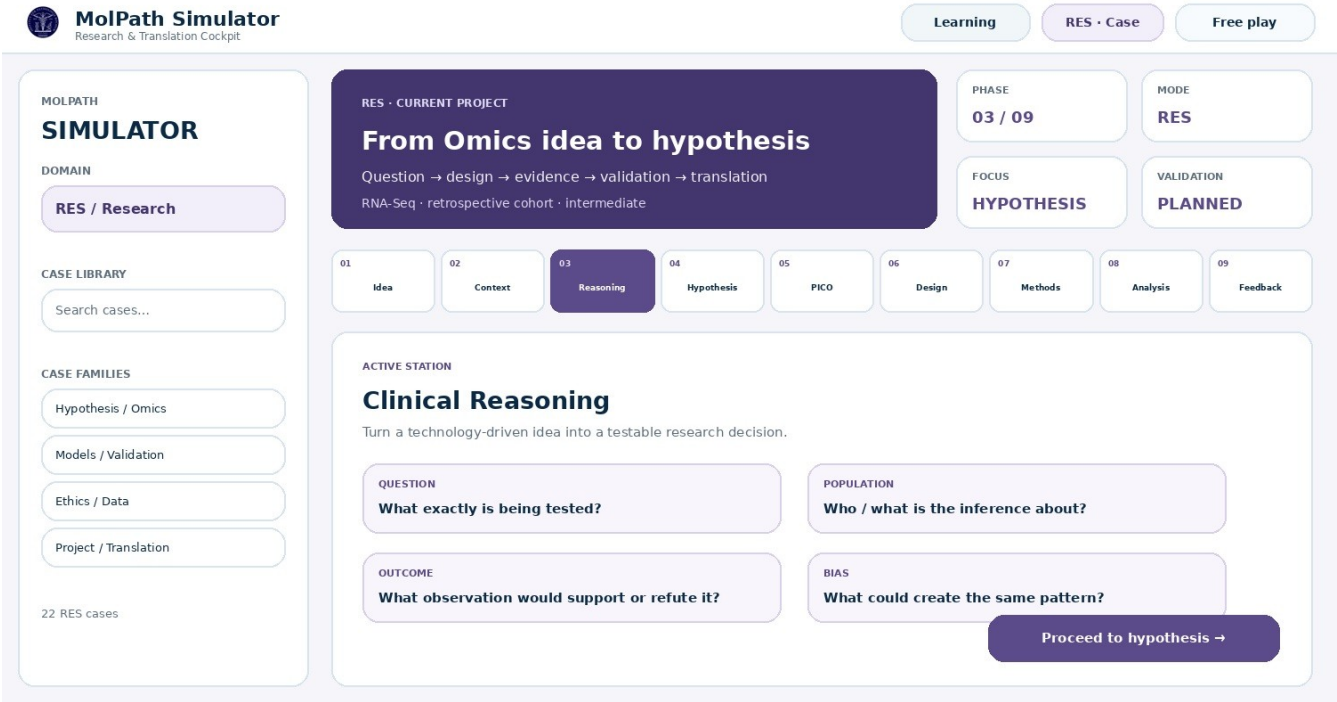
OPERATING PRINCIPLE	Define the question before the method. Separate discovery from confirmation. Plan bias control and validation before seeing the attractive result.
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CASE FAMILIES

RES / HYP Hypothesis · Omics · study planning From vague “measure everything” ideas to operational questions, PICO, endpoints, signatures and explicit exploratory/confirmatory boundaries.	RES / VAL Models · validation · preclinical bridge Model fitness, organoids, independent validation, orthogonal assays and the bridge from experimental effect to patient relevance.
RES / ETH Ethics · data · patient protection Consent, residual material, incidental findings, additional biopsies, data sharing, bias, transparency and governance.	RES / PM Project · team · translation Data/code structure, authorship, grants, interdisciplinary roles and the path from published biomarker to routine implementation.

The research cockpit

The RES interface keeps the simulator shell but changes the instrument logic. Instead of tissue budget or run-release state, the cockpit tracks project phase and evaluates five research dimensions: clinical question, hypothesis/PICO, study design, analysis plan, and translation/validation.



Schematic representation of the RES cockpit with the clinical-reasoning gate inserted into the current runtime profile.

COCKPIT SHIFT

MTB asks “What evidence changes the patient decision?”. LAB asks “Can this result/process be safely released?”. RES asks “What exactly can this study claim if the data look interesting?”.

Nine stations from project idea to defensible inference

01	02	03	04	05	06	07	08	09
Project Idea	Context	Reasoning	Hypothesis	PICO	Design	Methods	Analysis	Feedback

SCOPE 01–02 → GATE 03 → INFERENCE 04–05 → STUDY ARCHITECTURE 06–08 → FEEDBACK 09

STATION	PRIMARY OUTPUT	DO NOT PROCEED IF...
01 Project Idea	Problem worth investigating; provisional decision target	The project is only “we have a platform / cohort / biomarker”.
02 Context	Population, material, data, resources, ethics and constraints	Feasibility and governance are still assumptions.
03 Reasoning	Explicit research gap, inference target and key threats	You are choosing a hypothesis after seeing the attractive pattern.
04 Hypothesis	Specific, falsifiable and clinically/biologically meaningful proposition	The hypothesis is just a topic label or desired result.
05 PICO	Operational population, exposure/marker, comparator and outcome	Population, comparator, cutoff or endpoint remain undefined.
06 Design	Predefined primary/exploratory structure, cohorts, covariates and validation concept	Design cannot distinguish the target signal from obvious alternatives.
07 Methods	Methods fit the biological question, material and required resolution	Platform availability is the main justification.
08 Analysis	Analysis plan, multiplicity, confounders, QC and reproducibility are explicit	The analysis plan is “look at the heatmap and decide later”.
09 Feedback	Claim, limitations, validation/translation path and honest evidentiary level	The final story is stronger than the design can support.

Define the inference before generating the data

The most common RES failure mode is technology-led reasoning: the platform, dataset or sample collection becomes the project objective. The reasoning gate forces the learner to state what is unknown, what could answer it, and what alternative explanations must be controlled.

MISSION 01 · PROJECT IDEA What clinical or biological problem motivates the work? Which decision, mechanism or association would matter if supported?	BOUNDARY 02 · CONTEXT Who are the cases? What material and outcomes exist? Are data complete? Which ethics, consent, batch, resource or model constraints already shape the feasible question?
GATE 03 · REASONING What is the target inference? What observation would support or contradict it? Which confounders or biases could generate the same pattern?	LIMIT CLAIM CEILING Small, heterogeneous or retrospective datasets can be valuable - but they set a ceiling on confirmatory language and causal interpretation.

REASONING GATE QUESTIONS

- ☐ What exactly is the research question - not the platform or assay?
- ☐ What population or model is the intended inference about?
- ☐ Which primary outcome or biological readout would make the hypothesis testable?
- ☐ Which obvious confounder, batch effect, selection process or model mismatch could create a similar result?
- ☐ What must be predefined before access to outcome-dependent results?

DECISION GATE

Do not proceed to hypothesis selection until the project can be stated as a question with an inference target, a measurable outcome and a credible alternative explanation.

Hypothesis first. PICO makes it operational.

A useful hypothesis is specific enough to fail. PICO then converts the concept into analysable groups, measurements and outcomes. In biomarker research, “I” may be an exposure, molecular signature or marker state rather than an intervention.

P · Population Tumour entity, stage, treatment context, model system, eligibility and sampling frame.	I · Indicator / exposure Marker, pathway score, molecular alteration, intervention or experimentally defined state.
C · Comparator Reference group, alternative model, low marker state, responder/non-responder or matched control.	O · Outcome Clinical endpoint, response, survival, mechanistic readout or predefined quantitative phenotype.

EXPLORATORY IS VALID

Exploratory discovery is legitimate when labelled as such. The problem is not exploration; it is retroactively presenting an exploratory pattern, cutoff or subgroup as if it had been predefined.

NO-GO FORMULATIONS

- “We will sequence everything and see what becomes significant.”
- “The gene with the smallest p-value will be our biomarker.”
- “We will choose the cutoff after looking at the survival curves.”
- “All pathways that look interesting will be treated as primary findings.”

Design must make the intended claim possible

The design station is where the project’s claim ceiling is set. A study can be technically impeccable and still be unable to answer its stated question if cohort definition, controls, timing, confounders or validation are wrong.

STATION	PRIMARY OUTPUT	DO NOT PROCEED IF...
PRIMARY	Predefine the principal question, endpoint and main comparison.	Primary status depends on the most attractive post-hoc result.
EXPLORATORY	Declare discovery analyses, clustering, pathways and hypothesis generation explicitly.	Exploratory output is narrated as confirmatory evidence.
COHORT	Define inclusion/exclusion, sampling frame, treatment context and missing-data reality.	Availability alone defines the population.
CONFOUNDERS	Plan relevant covariates: tumour entity, stage, therapy, centre, batch, year, tumour fraction, model background.	Known alternative explanations are ignored because the assay is “objective”.
VALIDATION	Reserve an independent cohort, orthogonal assay or mechanistically distinct confirmation step.	Validation is only considered after the discovery result looks promising.
DESIGN TEST	Ask whether a negative result would still be interpretable. If “negative” could mean low power, poor material, wrong model, unmeasured confounding or failed assay, the design needs another control or limitation statement.	

RESOURCE DISCIPLINE

- Protect diagnostic reserve before consuming patient material for exploratory work.
- Match assay breadth to the hypothesis and validation budget; “more omics” is not automatically more informative.
- Plan roles, code/data structure and reproducible analysis before project complexity makes them emergency tasks.

Choose resolution deliberately; analyse what was designed

Methods and analysis are coupled decisions. The assay determines what signal can exist; the analysis determines what inference can be made from that signal. Neither step should be delegated to “whatever the platform” or “whatever the bioinformatics finds”.

ASSAY METHOD FIT Does the method measure the biological layer required by the hypothesis - DNA, RNA, protein, spatial context, cell state or functional phenotype?	INPUT MATERIAL FIT Is FFPE quality, tumour fraction, RNA integrity, sample size, model system or tissue consumption compatible with the proposed method?
PLAN ANALYSIS PLAN Specify QC, normalization, contrasts, covariates, multiple testing/FDR, missing data and software/code provenance before interpretation.	TRACE REPRODUCIBILITY Separate raw / processed data, code, environment/version information and README/metadata so the analysis can be reconstructed.

OMICS OPERATING LOGIC

- ☐ Gene list → pathway / network context → biologically coherent candidate or signature.
- ☐ Effect size and uncertainty matter alongside statistical significance.
- ☐ Multiple testing must be part of the plan, not a cosmetic adjustment after result selection.
- ☐ Batch and clinical covariates belong in the model when they can explain the observed contrast.
- ☐ Cutoffs and model complexity require protection against overfitting and optimistic internal performance.

CAUTION

A heatmap is a visualization, not a validation strategy. A low p-value is an analysis output, not a biological mechanism.

Land with the claim the study actually earned

The RES landing is not “a significant result”. It is a calibrated statement that connects the question, design, evidence and limitations to the next validation or translation step. The simulator therefore scores five dimensions rather than a single statistical outcome.

SCORE DIMENSION	WHAT GOOD PERFORMANCE LOOKS LIKE	TYPICAL LOSS OF CREDIT
Clinical question	Specific problem, population and decision/inference target.	Technology-led or vague project objective.
Hypothesis / PICO	Falsifiable proposition with operational comparison and outcome.	Topic labels, post-hoc cutoffs, no comparator.
Study design	Cohort/model, endpoints, controls, confounders and exploratory/primary boundary fit the claim.	Design cannot separate signal from obvious bias.
Analysis plan	QC, statistical/bioinformatic plan, multiplicity, covariates and reproducibility are explicit.	Data fishing, HARKing, undocumented analysis decisions.
Translation / validation	Independent/orthogonal validation and realistic implementation or mechanistic next step.	Discovery treated as final proof or routine-ready biomarker.
TRANSLATION LADDER	Discovery → technical/orthogonal confirmation → independent validation → clinical/biological utility → workflow, quality, cost and governance. Skipping rungs changes the strength of the claim, not merely the wording.	

Common research failure modes and recovery actions

ABNORMAL CONDITION	OPERATING RESPONSE	AVOID
"WE MEASURE EVERYTHING"	Return to question, population, endpoint and primary/exploratory split before consuming material.	Do not use platform availability as the scientific objective.
HARKING / POST-HOC CUTOFF	Relabel appropriately as exploratory; document how the hypothesis/cutoff was generated; require independent confirmation.	Do not rewrite the protocol history.
SINGLE-GENE OVERCALL	Move from isolated hit to pathway/network coherence, effect size, technical validity and external/orthogonal confirmation.	Do not equate smallest p-value with mechanism.
MODEL MISMATCH	Reassess whether cell line/organoid/model represents the tumour biology and treatment context needed for the claim.	Do not generalize model-specific effects directly to patients.
GOVERNANCE GAP	Stop before further use where consent, ethics, material release, incidental findings or data-sharing rules are unresolved.	Do not treat residual FFPE as freely available research material.
NO VALIDATION PATH	Reduce claim strength and design a realistic technical, independent or patient-level validation step.	Do not call discovery "validated" because it is internally reproducible.
CODE / DATA CHAOS	Freeze raw data, establish processed/code/README structure, version the environment and document analysis provenance.	Do not continue creating final_neu_v7_END files.
PRIORITY	Protect interpretability before novelty. A smaller, predefined, well-governed project can support a stronger conclusion than a broader dataset assembled without an inference plan.	

Take-off checklist

Use before committing resources, material or outcome-dependent analysis. The checklist is front-loaded because the cheapest time to prevent HARKing, model mismatch, governance gaps and irreproducible analysis is before data generation.

MISSION & QUESTION ☐ Research problem stated without naming the technology as the objective ☐ Target population / model and inference target defined ☐ Primary outcome / biological readout identified ☐ Exploratory vs confirmatory intent declared	DESIGN & GOVERNANCE ☐ Inclusion / comparator / timing and key confounders defined ☐ Material and model fitness checked ☐ Ethics / consent / data / incidental-finding obligations cleared ☐ Independent or orthogonal validation route planned
METHODS & ANALYSIS ☐ Assay resolution matches the biological question ☐ QC, normalization, contrasts and missing-data handling planned ☐ Multiplicity / FDR and cutoff strategy specified ☐ Raw / processed / code / version provenance established	CLAIM & TRANSLATION ☐ Expected claim matches design strength ☐ Alternative explanations and limitations stated ☐ Validation is not confused with internal repetition ☐ Next mechanistic / clinical / implementation step is realistic

TAKE-OFF
CLEARANCE

Proceed when the study can be described as a testable inference with a defensible design, planned analysis, governance clearance and a validation path.

PROJECT _____ DATE _____ CLEARED ☐

The three operational cockpits are complete. Section 4 adds the current v2.6.0-rc1 Methods Focus architecture as a cross-domain decision layer rather than a fourth cockpit.

Methodology is a second axis, not a fourth cockpit

The current v2.6.0-rc1 simulator keeps MTB, LAB and RES as the three operating environments and overlays an independent Methods focus filter. A case belongs here only when the methodological decision materially determines the correct resolution - not merely because a molecular or analytical technique appears somewhere in the case.

DRIVER CASES		FOCUS DOMAINS		OPERATING AREAS	
44 / 91		5 non-exclusive		MTB · LAB · RES	
OPERATING PRINCIPLE		Choose the cockpit by role and decision context. Add Methods Focus when assay choice, sample suitability, analytical validity, computational design or methodological limitations are themselves part of the decision.			
PRIMARY AXIS		SECOND AXIS			
MTB · LAB · RES		Methods Focus			
Where am I operating, and what type of decision am I responsible for?		Which methodological pitfall or design constraint materially drives this case?			
INTERSECTION		CURATION			
Filters combine		Driver ≠ method present			
Methods Focus intersects with area, search, difficulty and Signature filters rather than replacing them.		A method can be present without qualifying the case. Eligibility is based on decision impact, not technique occurrence.			

Current strict curation: MTB 14 · LAB 16 · RES 14 method-driven cases. 18 cases carry more than one focus domain.

One case can be clinical, laboratory or research - and method-driven at the same time

The two axes answer different questions. Primary mode identifies the operational environment. Methods Focus identifies the methodological reasoning that deserves explicit attention. The same case therefore keeps its MTB, LAB or RES workflow while receiving one or more focus tags.

AXIS	QUESTION	VALUES	WHAT IT CHANGES
PRIMARY MODE	Where am I operating?	MTB · LAB · RES	Workflow, station sequence, landing decision and score context.
METHODS FOCUS	What methodological issue drives the resolution?	5 focus domains	Case discovery, cross-domain teaching emphasis and methodological cross-check.
OTHER FILTERS	Which subset do I want now?	Search · difficulty · Signature	Intersects with both axes; does not redefine case identity.
MTB METHODS DRIVER		LAB METHODS DRIVER	RES METHODS DRIVER
14 cases		16 cases	14 cases
ARCHITECTURE NOTE	Historical MET case IDs are retained for compatibility, but they no longer define the filter. Eight of the ten original MET-labelled cases remain method drivers; two are deliberately excluded because their decisive challenge is evidence prioritisation / interpretation rather than a methodological failure or choice.		

Method occurrence alone does not qualify a case

The strict audit distinguishes method-driven cases from cases in which a technique is only supporting context. This prevents the filter from becoming a synonym for “contains NGS / PCR / IHC / statistics”.

ROLE	INCLUDE?	OPERATIONAL DEFINITION
DRIVER	YES	The correct resolution depends materially on method selection, sample fitness, analytical validity, sensitivity, QC, computational design, validation or a directly related limitation.
SUPPORTING	NO	A method contributes information, but the core challenge is biological, clinical, evidentiary, ethical, communicative or administrative.
NONE	NO	Methodology is incidental background and does not change the key decision.
MULTI-DOMAIN	YES	A driver case may carry more than one focus tag when distinct methodological layers jointly matter.
LEGACY MET ID	NOT AUTOMATIC	Identifier compatibility is preserved; inclusion is determined by the same driver rule used for all MTB / LAB / RES cases.
DRIVER EXAMPLE MET_NGS_001 · coverage gap A therapy-relevant exon is inadequately covered; the negative result cannot be interpreted without addressing the analytical blind spot.	DRIVER EXAMPLE LAB_POST_001 · pipeline change Change control, validation and reproducibility directly determine whether current results remain defensible.	
DRIVER EXAMPLE RES_HYP_001 · “measure everything” Exploratory vs confirmatory design and hypothesis definition determine whether downstream inference is valid.	SUPPORTING EXAMPLE MET_NGS_003 · VUS The decisive problem is evidence/actionability interpretation; the assay itself is not the methodological bottleneck.	
NO-GO	Do not add a case to Methods Focus merely because a technique is visible in the workflow. The learner must have to reason about the method, its validity or its inferential limits to resolve the case correctly.	

Five lenses, deliberately non-exclusive

Focus tags describe the methodological layer that carries the decision. Counts overlap because a single case can require more than one kind of methodological reasoning.

FOCUS	CURRENT CASES	TYPICAL DRIVER ISSUES
Molecular diagnostics	20	Assay selection · coverage · fusion detection · sensitivity / LoD · negative-result limits · orthogonal testing
Preanalytics · analytics · QC	18	Preanalytics · material adequacy · controls · contamination · run acceptance · validation / verification · release validity
Research & experimental design	12	Hypothesis design · exploratory/confirmatory boundary · operationalisation · model selection · independent validation · translation
Bioinformatics & statistics	10	Pipeline validation · change control · omics prioritisation · cutoffs · multiple testing · version control · reproducibility
Digital pathology & AI	3	Image quality · local / algorithm validation · training bias · generalisability · human review
INTERPRETATION	The five counts sum to more than 44 because 18 current driver cases are intentionally multi-domain. A case should carry the smallest set of tags that explains the actual methodological decision.	

Six questions before you trust the method

Use this cross-check inside any MTB, LAB or RES case whenever the method itself becomes decision-critical. It complements the cockpit-specific checklist; it does not replace it.

01 · QUESTION What exactly must the method answer? State the information gap and the decision that a positive, negative or indeterminate result could change.	02 · INPUT Is the specimen, dataset, image or model fit for this question? Check identity, representativeness, quantity, quality and relevant preanalytical history.
03 · METHOD Does the selected assay / design measure the required biological or analytical layer at the necessary resolution and sensitivity?	04 · VALIDITY Are controls, coverage, QC, pipeline state, validation status and analysis assumptions sufficient to support interpretation?
05 · INTERPRETATION What does the result establish - and what does it not exclude? Keep analytical scope separate from biological certainty.	06 · ACTION Report, release, repeat, confirm orthogonally, obtain new input, redesign, validate further - or stop because more testing adds no decision value.
DECISION GATE	If you cannot explain how the methodological result changes the next step, do not commit additional material, run time, compute or interpretive weight yet.
NO-GO	Broad testing is not a substitute for an undefined question. Technical sophistication cannot rescue unsuitable input, invalid QC, an unvalidated pipeline or an inference the design was never built to support.

A negative result is only as broad as the validated path that produced it

Read every result through four validity layers. Failure at an earlier layer constrains all downstream interpretation, even when the final report or plot looks plausible.

1	2	3	4
INPUT FIT	METHOD FIT	EXECUTION	INFERENCE
Correct input and adequate quality?	Correct analyte, resolution and sensitivity?	VALID Controls, QC, coverage and pipeline acceptable?	ALLOWED Does the result support the claim?
FINDING	WHAT IT MAY MEAN	WHAT IT DOES NOT AUTOMATICALLY MEAN	NEXT MOVE
NEGATIVE + VALID	Target not detected within validated scope and sensitivity.	Biology absent under all conditions / sample sites / analyte classes.	Report scope and residual uncertainty explicitly.
NEGATIVE + BLIND SPOT	No call in the observed portion.	Target excluded.	Resolve coverage / analyte / sensitivity gap or use orthogonal / alternative testing.
QC / INPUT INVALID	Result may be technically uninterpretable.	“Probably negative” because the trace / morphology / prior result looks plausible.	Hold interpretation; repeat from the process step that addresses the suspected failure.
CAUTION	Repeat testing can correct a result without explaining the failure mechanism. In LAB and RES especially, distinguish result recovery from process / model / pipeline validation.		

Same methods logic, different landing

Methods Focus never changes the cockpit's operational endpoint. It changes what must be reasoned through on the way to that endpoint.

COCKPIT	METHOD-DRIVEN EXAMPLE	METHODS QUESTION	LANDING
MTB	Fusion suspicion despite negative DNA testing	Is the assay able to detect the alteration class with sufficient sensitivity, or is RNA / FISH / another strategy required?	A patient-care testing / interpretation recommendation.
LAB	Hotspot signal in the negative control	Is the run valid, what is the contamination scope, and which evidence separates local from batch-wide failure?	Release / hold / repeat decision plus deviation / CAPA path.
RES	RNA-seq yields hundreds of DEGs	Which predefined analysis, prioritisation and validation strategy turns a large signal space into a defensible inference?	A calibrated research claim and next validation step.
OPERATING PRINCIPLE	Do not move the case into a generic “methods workflow”. Stay inside MTB, LAB or RES and apply the Methods Focus cross-check exactly where the methodological constraint becomes decision-critical.		

Common methodological failure modes and recovery actions

ABNORMAL CONDITION	OPERATING RESPONSE	AVOID
METHOD CHOSEN BECAUSE AVAILABLE	Return to the information gap; compare assay scope, sensitivity, input needs and decision impact before committing.	Do not let platform availability define the question.
INPUT NOT FIT FOR PURPOSE	Reassess identity, representativeness, quantity, quality and preanalytics; obtain alternative input or explicitly limit the claim.	Do not compensate for poor input by over-interpreting sophisticated output.
CONTROL / QC FAILURE	Hold interpretation or release; define affected scope and resolve the validity failure before using patient / project results.	Do not treat plausible-looking results as independent evidence of validity.
NEGATIVE OVERCLAIM	State analytical scope, LoD / power / coverage and relevant sampling limitations; use orthogonal or alternative testing when decision-relevant.	Do not translate “not detected” into “biologically absent” without qualification.
UNVALIDATED PIPELINE / CHANGE	Document version state, assess impact, validate / verify the change and reanalyse affected data when required.	Do not assume computational changes are neutral because raw data are unchanged.
POST-HOC ANALYSIS DRIFT	Re-establish predefined vs exploratory analyses, control multiplicity / overfitting and require independent validation for selected findings.	Do not relabel discovery as confirmation after seeing the outcome.
DIGITAL / AI OUTPUT WITHOUT VALIDATION	Check image / input quality, local performance, generalisability and human-review responsibility before operational use.	Do not treat an algorithmic score as self-validating evidence.
PRIORITY	Protect validity before completeness. A narrower defensible result is better than a comprehensive output whose input, method, QC or inference limits are unresolved.	

Pre-commit cross-check

Use whenever methodology materially drives the case. Complete the relevant MTB / LAB / RES take-off checklist as usual, then use this page to test method fit, validity and inferential scope before committing.

QUESTION & IMPACT ☐ Information gap stated explicitly ☐ Positive / negative / indeterminate result linked to a next decision ☐ Method is being selected for the question, not because it is available	INPUT & METHOD FIT ☐ Identity / representativeness / quantity / quality adequate ☐ Preanalytics / model / dataset provenance reviewed ☐ Assay / design matches biological layer, resolution and sensitivity
VALIDITY & INTERPRETATION ☐ Controls / QC / coverage / pipeline / assumptions acceptable ☐ Negative-result limits or statistical power understood ☐ Result statement stays within validated analytical and inferential scope	ACTION & HANDOFF ☐ Report / release / repeat / orthogonal test / redesign option selected deliberately ☐ Further testing has defined decision value ☐ Residual uncertainty and required follow-up handed off explicitly
DECISION GATE	Can I defend the input, method, validity and inference - and explain exactly what operational decision this result supports?
NO-GO	Do not commit to an interpretation while a known input, QC, coverage, validation, model-fit or analysis limitation could materially change the conclusion.
CASE / PROJECT _____ DATE _____ CLEARED ☐	

Manual architecture complete: MTB Operations · LAB Operations · RES Operations · Methods Focus. Methods Focus remains an orthogonal learning layer across the three cockpits.